

A. one leaf spring
Rayleigh shape $\varphi \rightarrow k, m$

B. two dipoles
 $U(r) \rightarrow F(r), \gamma = -F'(d)$

C. Lagrangian $m\ddot{x}_i = -kx_i \mp F$, equilibrium, linearise

extension:
keep r^{-4}
exactly

E. damping
same λ both modes

D. normal modes η, ξ decouple

F. release
 η_0, ξ_0 , beats

KEY RESULT $f_2^2 - f_1^2 = \frac{\gamma}{2\pi^2 m} = C d^{-5}, C = \frac{3\mu_0 \mu_m^2}{\pi^3 m}$

measurables: two FFT peaks at f_1, f_2 · peak heights $\xi_0/|\eta_0| = (f_1/f_2)^2$ · beat period $1/(f_2 - f_1)$ · equal decay